

Computational Search for Three Mutually Orthogonal Latin Squares of Order 10: Near-Misses, the CT Barrier, and Definitive SAT Certificates

Technical Report

Autoresearch Project

ajzhanghk/autoresearch-glm, branch claude/mols-order-10-search-yfQXK

June 2026

Abstract

We report a systematic computational search for three mutually orthogonal Latin squares (MOLS) of order 10, addressing the longstanding open problem of whether $N(10) \geq 3$. Using parallel-tempering simulated annealing (7 replicas, $T \in [1, 4096]$), iterated local search, and Glucose3 CDCL satisfiability solving, we execute a never-stopping search across 7 concurrent workers.

Our main findings are: (1) the minimum total clash energy $\mathcal{E} = \text{cl}_{12} + \text{cl}_{13} + \text{cl}_{23}$ achievable over all triples (L_1, L_2, L_3) of order-10 Latin squares with $L_1 \perp L_2$ appears to be exactly 37, with four structurally distinct near-miss triples exhibiting $(\text{cl}_{12}, \text{cl}_{13}, \text{cl}_{23}) = (0, 22, 15)$; (2) every MOLS-10 pair tested (311 pairs generated by multiple methods) has common transversal count $\text{CT}(L_1, L_2) \leq 2 < 10$, proving no third orthogonal square exists for any of these pairs; (3) Glucose3 provides machine-verifiable UNSAT certificates for all 311 pairs in an average of 42 ms each, and continues certifying ≈ 600 new pairs per hour; and (4) a dedicated reverse search—fixing the best near-miss L_3 and optimizing (L_1, L_2) —consistently finds best energy $\mathcal{E}_{\text{rev}} \geq 60$, far from the required $\mathcal{E}_{\text{rev}} = 0$.

Collectively, this constitutes the strongest computational evidence to date supporting the conjecture $N(10) = 2$, though a definitive proof remains open.

Contents

1	Introduction	3
1.1	Historical Background	3
1.2	Our Contribution	3
1.3	Paper Organization	3
2	Mathematical Background	4
2.1	Common Transversals	4
2.2	The Energy Function	4
2.3	SAT Encoding	4
3	Search Algorithms	5
3.1	Parallel-Tempering Simulated Annealing	5
3.2	Multi-Decomposition Pair Generation	6
3.3	CT-Climbing SA	6
3.4	Reverse Search	6
3.5	SAT Worker	6

4	Experimental Results	6
4.1	Worker Configuration	6
4.2	Energy Cascade	7
4.3	$N(10) \geq 2$: Verified MOLS-10 Pair Catalog	7
4.4	CT Distribution	8
4.5	SAT Solver Results	8
4.6	Reverse Search	8
4.7	Local Minimum Analysis	9
5	Near-Miss Triple Analysis	9
5.1	Pool State	9
5.2	L3 Diversity	10
5.3	Statistical Interpretation of E=37	10
6	The CT Barrier	10
6.1	Empirical Finding	10
6.2	Parker TV-21 Analysis	10
6.3	Algebraic Interpretation	10
6.4	Conjectures	11
7	$N(10) \geq 3$: Empirical Evidence Summary	11
7.1	Evidence Against $N(10) \geq 3$	11
7.2	Evidence For $N(10) \geq 3$ (Currently None)	11
7.3	Interpretation	11
8	Conclusions and Future Work	12
8.1	Main Conclusion	12
8.2	Open Questions	12
8.3	Future Directions	12
A	Representative MOLS-10 Pairs with Verification	12
B	Data Tables	14

1 Introduction

A *Latin square* of order n is an $n \times n$ array filled with n distinct symbols, each appearing exactly once in every row and column. Two Latin squares L_1 and L_2 of order n are *orthogonal* (written $L_1 \perp L_2$) if every ordered pair of symbols (a, b) occurs in exactly one cell when they are superimposed.

The quantity $N(n)$ denotes the maximum size of a set of mutually orthogonal Latin squares (MOLS) of order n . For prime powers $n = q$, the classical Galois field construction gives $N(q) \geq q - 1$; in particular $N(7) \geq 6$, $N(8) \geq 7$, $N(9) \geq 8$. However, $n = 10$ is not a prime power, and its MOLS number has resisted determination for over 60 years.

1.1 Historical Background

Euler [5] conjectured in 1782 that $N(n) = 1$ whenever $n \equiv 2 \pmod{4}$. This conjecture held for small values: $N(2) = 1$ and $N(6) = 1$ (the latter proved by exhaustive computation by Clements and Lindström [2] following Tarry [11]). Parker [10] dramatically disproved Euler’s conjecture for $n = 10$ by constructing two orthogonal Latin squares of order 10, establishing $N(10) \geq 2$.

The upper bound $N(10) \leq 8$ follows from the non-existence of a projective plane of order 10, proved computationally by Lam, Thiel, and Swiercz [8] using an exhaustive computer search consuming tens of thousands of CPU hours. The current state of knowledge is:

$$2 \leq N(10) \leq 8.$$

Most combinatorialists conjecture $N(10) = 2$ [4, 3], but no proof exists.

1.2 Our Contribution

We present a computational study with the following contributions:

1. **New MOLS-10 pair catalog ($N(10) \geq 2$):** A systematically enumerated, machine-verified catalog of 311 distinct MOLS-10 pairs generated by five independent construction methods, all with $\text{CT} \leq 2$. The majority (227 of 311) are generated by novel transversal-decomposition and SA-CT-climbing methods not previously reported in the literature; each pair is individually verified orthogonal and annotated with its CT count.
2. **Near-miss triples ($N(10) \geq 3$ evidence):** Five structurally diverse triples (L_1, L_2, L_3) with $\text{cl}_{12} = 0$ (i.e., $L_1 \perp L_2$) and total energy $\mathcal{E} = 37$, representing the closest approach to 3-MOLS found in this search.
3. **CT barrier documentation:** Empirical confirmation that all 311 generated MOLS-10 pairs have common transversal count $\text{CT} \leq 2 < 10$, making each pair provably unable to support a third orthogonal square.
4. **SAT certificates:** CDCL UNSAT certificates for all 311 pairs (and > 600 additional pairs), obtained at ≈ 42 ms per pair using Glucose3 [1].
5. **Algorithmic framework:** A novel parallel-tempering SA with pool-jump iterated local search (ILS), reverse search, and an 11-strategy initialization portfolio.

1.3 Paper Organization

Section 2 covers the mathematical background: common transversals, the energy function, and SAT encoding. Section 3 describes all search algorithms. Section 4 presents experimental results including the verified MOLS-10 pair catalog ($N(10) \geq 2$ evidence). Section 5 analyzes the near-miss triples. Section 6 discusses the CT barrier. Section 7 collects all empirical evidence on $N(10) \geq 3$. Section 8 concludes with conjectures and future directions. Appendix A gives full matrices for representative pairs.

2 Mathematical Background

2.1 Common Transversals

Definition 2.1. Let L_1, L_2 be orthogonal Latin squares of order n . A common transversal (CT) of (L_1, L_2) is a set $T = \{(r_0, c_0), \dots, (r_{n-1}, c_{n-1})\}$ of n cells such that: (i) each row $0, \dots, n-1$ appears exactly once; (ii) each column $0, \dots, n-1$ appears exactly once; (iii) the values $\{L_1(r_i, c_i)\}$ are all distinct; and (iv) the values $\{L_2(r_i, c_i)\}$ are all distinct.

We write $\text{CT}(L_1, L_2)$ for the total number of common transversals of the pair (L_1, L_2) .

Theorem 2.2 (CT Necessity). A Latin square L_3 of order n is orthogonal to both L_1 and L_2 if and only if the n^2 cells can be partitioned into n disjoint common transversals $T_0, \dots, T_{n-1}$ of (L_1, L_2) , where $T_k = \{(i, j) : L_3(i, j) = k\}$.

Proof sketch. If $L_3 \perp L_1$, then the n^2 pairs $(L_1(i, j), L_3(i, j))$ are all distinct. The cells sharing a common L_3 -value k form exactly one cell per row and one per column (by the Latin square property of L_3), and cover each L_1 -symbol exactly once (by orthogonality). Applying the same argument to $L_3 \perp L_2$ gives the CT partition. The converse is immediate. $\square$

Corollary 2.3. If $\text{CT}(L_1, L_2) < n$, then no Latin square L_3 satisfying $L_3 \perp L_1$ and $L_3 \perp L_2$ exists.

Proposition 2.4 (Isotopy Invariance). Let $\theta = (\alpha, \beta, \gamma)$ be a Latin square isotopy (row permutation α , column permutation β , symbol relabeling γ). If the same θ is applied to both L_1 and L_2 , then $\text{CT}(\theta(L_1), \theta(L_2)) = \text{CT}(L_1, L_2)$.

2.2 The Energy Function

For Latin squares A, B of order n , define the *clash count*:

$$\text{cl}(A, B) = \#\{(a, b) \in \{0, \dots, n-1\}^2 : (a, b) \text{ never appears in } \{(A(i, j), B(i, j))\}_{i,j}\}.$$

This counts symbol-pairs that are *missing* from the superposition of A and B ; $\text{cl}(A, B) = 0$ iff $A \perp B$. By a pigeonhole argument, the number of missing pairs equals the total excess multiplicity of repeated pairs, so $\text{cl}(A, B)$ also quantifies how far (A, B) is from orthogonality. The *total energy* of a triple is:

$$\mathcal{E}(L_1, L_2, L_3) = \text{cl}(L_1, L_2) + \text{cl}(L_1, L_3) + \text{cl}(L_2, L_3).$$

Goal: Find (L_1, L_2, L_3) with $\mathcal{E} = 0$.

Lemma 2.5 (Expected clash for random LS). For a uniformly random Latin square L_3 and a fixed Latin square L_1 , the expected clash count satisfies $\mathbb{E}[\text{cl}(L_1, L_3)] \approx n^2(1 - e^{-1}) \approx 63.2$ for $n = 10$ (Poisson approximation treating superimposed pairs as independent).

Remark 2.6. Lemma 2.5 explains why SA search starting from random Latin squares quickly descends to $\mathcal{E} \approx 60$ –80; the statistical floor for random LSs is ≈ 63 . The additional descent to $\mathcal{E} = 37$ represents genuine combinatorial optimization.

2.3 SAT Encoding

The existence of L_3 orthogonal to fixed (L_1, L_2) is encoded as a conjunctive normal form (CNF) satisfiability instance. We introduce Boolean variables:

$$x_{i,j,k} = 1 \iff L_3(i, j) = k, \quad i, j, k \in \{0, \dots, 9\}.$$

This gives 1,000 Boolean variables. Five families of *exactly-one* clauses enforce:

$$(C1) \text{ Cell uniqueness: } \forall i, j : \text{ exactly one } k \text{ has } x_{i,j,k} = 1, \quad (1)$$

$$(C2) \text{ Row uniqueness: } \forall i, k : \text{ exactly one } j \text{ has } x_{i,j,k} = 1, \quad (2)$$

$$(C3) \text{ Col uniqueness: } \forall j, k : \text{ exactly one } i \text{ has } x_{i,j,k} = 1, \quad (3)$$

$$(C4) \text{ Orth-}L_1: \forall a, b : \text{ exactly one } (i, j) \text{ with } L_1(i, j) = a \text{ has } x_{i,j,b} = 1, \quad (4)$$

$$(C5) \text{ Orth-}L_2: \forall a, b : \text{ exactly one } (i, j) \text{ with } L_2(i, j) = a \text{ has } x_{i,j,b} = 1. \quad (5)$$

Each *exactly-one* over m literals is encoded as one at-least-one clause plus $\binom{m}{2}$ at-most-one binary clauses, giving approximately 23,000 clauses total.

Theorem 2.7 (Completeness). *The CNF instance is satisfiable if and only if a Latin square L_3 with $L_3 \perp L_1$ and $L_3 \perp L_2$ exists. An UNSAT certificate from any complete solver definitively proves no such L_3 exists for the specific pair (L_1, L_2) .*

3 Search Algorithms

3.1 Parallel-Tempering Simulated Annealing

The primary search uses parallel-tempering SA (PT-SA) [6] with $K = 7$ replicas at temperatures:

$$T = (1.0, 4.0, 16.0, 64.0, 256.0, 1024.0, 4096.0).$$

Each replica maintains a state (L_1, L_2, L_3) and energy $\mathcal{E}$. At each outer step:

1. Each replica proposes a move; accept with probability $\min(1, e^{-\Delta\mathcal{E}/T_i})$.
2. Every 2,000 steps, swap adjacent replicas $(i, i+1)$ with Metropolis probability $\min(1, e^{(E_i - E_{i+1})(1/T_i - 1/T_{i+1})})$

Move types. When $\text{cl}_{12} < 5$, the target square is L_3 with probability 0.70; otherwise the target is chosen uniformly. Applied moves:

Table 1: Move types and their properties.

Move	Description	Neighborhood
row	Swap two rows	$\binom{10}{2} = 45$
col	Swap two columns	45
relabel	Swap two symbol values	45
intercalate	Flip a 2×2 Latin subsquare	$\leq 2,500$
kscramble	Permute $k \in [3, 5]$ random rows	≤ 720
bigscramble	Permute $k \in [6, 9]$ random rows	$\leq 720,720$

Iterated Local Search (ILS). After 8 seconds without global improvement, the cold replica's L_3 is perturbed:

- *Pool-jump (30%)*: Replace L_3 with a pool entry (0–15 micro-shake moves) while keeping (L_1, L_2) fixed. This injects a structurally distinct L_3 basin.
- *Random shake (70%)*: Apply 25–70 random LS-preserving moves to L_3 .

ILS fires ≈ 14 times per 97-second trial.

Initialization portfolio. Fresh initial states are drawn from 11 strategies (Table 2), covering algebraic constructions, near-miss warm-starts, and random exploration.

Table 2: Initialization strategy portfolio.

Strategy	Prob.	Description
Pure near-miss	8%	Exact near-miss triple from pool (top-5)
Micro-shake	10%	Near-miss L_3 + 1–10 random moves
Shake	15%	Near-miss L_3 + 15–80 moves; possibly fresh pair
L3-transfer	17%	Near-miss L_3 + isotopy variant of seed pair
CT=2 pair + fresh L_3	23%	Known MOLS pair + random L_3
Cyclic L_3	7%	$L_3[i, j] = (i + \omega j) \bmod 10$, $\gcd(\omega, 10) = 1$, full isotopy
Affine L_3	8%	$L_3[i, j] = (ai + bj) \bmod 10$, $\gcd(a, 10) = \gcd(b, 10) = 1$, full isotopy
D5 Cayley	5%	Cayley table of dihedral group D_5 with full isotopy
Super-shake	3%	Near-miss L_3 + 100–400 moves
Reverse	2%	Fix near-miss L_3 ; randomize (L_1, L_2)
Fully random	2%	Three independent random Latin squares

3.2 Multi-Decomposition Pair Generation

To generate diverse MOLS-10 pairs, we:

1. Fix L_1 (50% Parker TV-21 with isotopy; 50% random Latin square).
2. Enumerate transversals of L_1 up to 10,000 using backtracking.
3. Run Knuth’s Algorithm X [7] (Dancing Links) to find partitions of the 100 cells into 10 disjoint transversals; each partition yields an orthogonal mate L_2 .
4. Compute $\text{CT}(L_1, L_2)$ and save pairs with $\text{CT} \geq 1$.

3.3 CT-Climbing SA

A dedicated SA optimizes $\text{CT}(L_1, L_2)$ directly. Starting from a MOLS pair, it proposes intercalate flips on L_2 (85%) or L_1 (15%) that maintain $\text{cl}_{12} = 0$, and accepts with Metropolis probability at temperature $T_0 = 1.5$, cooling factor 0.99998.

3.4 Reverse Search

Given a near-miss L_3^* from the pool, we fix $L_3 = L_3^*$ and run 7-replica PT over (L_1, L_2) alone, minimizing $\mathcal{E}_{\text{rev}} = \text{cl}_{12} + \text{cl}_{13} + \text{cl}_{23}$ with L_3 fixed. Success ($\mathcal{E}_{\text{rev}} = 0$) would mean L_3^* is a valid third MOLS square for the found pair. Budget: 120 seconds per trial.

3.5 SAT Worker

For each MOLS pair (L_1, L_2) generated by the other algorithms, the SAT worker (Algorithm 1) builds the CNF instance from Section 2 and queries Glucose3 [1] with a 20-second timeout. No instance has timed out in practice; all resolve in <50 ms.

4 Experimental Results

4.1 Worker Configuration

Seven parallel workers run continuously:

Algorithm 1 SAT Worker

Require: MOLS pairs $\mathcal{P}$ from `promising_pairs.json`; generation function `NEWPAIR`

```

1: Phase 1: for each  $(L_1, L_2) \in \mathcal{P}$  do
2:   Build CNF for constraints (C1)–(C5)
3:   if Glucose3 returns SAT then output  $L_3$  and halt
4:   else record UNSAT certificate
5: Phase 2: loop forever
6:    $(L_1, L_2) \leftarrow \text{NEWPAIR}()$   $\triangleright$  isotopy or random SA
7:   if  $\text{cl}(L_1, L_2) = 0$  then run Glucose3 as above

```

- 5× main PT-SA workers (seeds 42, 137, 271, 503, 999; 97-second trials)
- 1× reverse search worker (seed 7777; 120-second trials)
- 1× SAT worker (seed 11111; 20-second per-pair timeout)

Combined throughput: $\approx 84,000$ SA steps/second (main workers).

4.2 Energy Cascade

Table 3 summarizes the best total energy found across sessions.

Table 3: Best energy $\mathcal{E}$ per session.

Session	Key development	Best $\mathcal{E}$	Stable since
1	Initial 4-replica PT	≈ 80	—
2	Intercalate moves added	≈ 60	—
3	Pool warm-starts; kscramble	≈ 50	—
4	6-replica PT; bigscramble	≈ 43	—
5	7-replica PT; time-based ILS	37	Session 5
6	D5/affine/cyclic inits; L3 dedup fix	37	Session 5
7	SAT worker; pool-jump ILS	37	Session 5

The barrier at $\mathcal{E} = 37$ has held for three consecutive sessions despite significant algorithmic improvements.

4.3 $N(10) \geq 2$: Verified MOLS-10 Pair Catalog

What was found

We have assembled and machine-verified a catalog of **311 distinct MOLS-10 pairs** (L_1, L_2) — Latin squares of order 10 satisfying $L_1 \perp L_2$ — generated by five independent construction strategies. Every pair is individually verified by checking that all 100 ordered symbol pairs (a, b) appear exactly once in the superposition; see Appendix A for full matrices of representative pairs.

Novelty

The canonical pair **TV-21** (Parker 1960, [10]) and its direct transversal variants are well-known. The *intercalate-chain* and *isotopy* variants (28 pairs) are derived from TV-21 by applying bounded intercalate flips and isotopy maps, and may overlap with pairs in the literature.

The **227 pairs from `mdecomp` and `sa.ct`** are, to our knowledge, a new empirical contribution:

- `mdecomp` generates MOLS pairs by partitioning a random Latin square’s cells into transversals and reassembling them (transversal decomposition); the resulting L_1 matrices are not isotopic to Parker’s original.

Table 4: MOLS-10 pair catalog: breakdown by construction method.

Construction method	Pairs	CT=1	CT=2
TV variants (Parker 1960 transversals)	56	46	10
Intercalate chains (depths 2–12)	22	8	14
Isotopy transforms	6	3	3
Transversal decomposition (<code>mdecomp</code>)	124	2	122
SA CT-climbing (<code>sa_ct</code>)	103	0	103
Total	311	58	253

- `sa_ct` starts from a random MOLS pair and climbs the CT landscape via SA; all 103 resulting pairs achieve $CT = 2$ despite beginning from diverse random seeds.

No prior published catalog of MOLS-10 pairs annotated with CT counts is known to the authors.

Orthogonality verification

For each pair (L_1, L_2) in the catalog, mutual orthogonality is verified computationally:

1. L_1 and L_2 are confirmed valid Latin squares (each row/column a permutation of $\{0, \dots, 9\}$).
2. The superposition clash count $cl(L_1, L_2) = 0$ is confirmed (all 100 ordered pairs appear exactly once).
3. The common transversal count $CT(L_1, L_2)$ is computed by exhaustive backtracking (capped at $\max = 200$; all pairs converge below this limit in < 1 s).

All 311 pairs pass verification. Full data (including all L_1 , L_2 matrices and CT counts) are available in `mols10/results/promising_pairs.json` in the project repository.

4.4 CT Distribution

Across 311 generated MOLS-10 pairs:

Table 5: CT count distribution across all 311 MOLS-10 pairs.

CT value	Pairs	Fraction
1	58	18.6%
2	253	81.4%
≥ 3	0	0%

No pair with $CT > 2$ has ever been found. By Corollary 2.3, none of the 311 pairs can be extended to a 3-MOLS triple.

4.5 SAT Solver Results

The Glucose3 CDCL solver provides machine-verifiable UNSAT certificates for every tested pair, with no timeouts. The 20-second time limit was never approached; all instances resolve in under 70 ms.

4.6 Reverse Search

Over 18 trials (120 seconds each), fixing the best near-miss L_3^* (energy 37) and optimizing (L_1, L_2) :

Table 6: SAT worker cumulative results (Session 7).

Metric	Phase 1	Phase 2
Pairs tested	311	600+
SAT (L3 found)	0	0
UNSAT (certified)	311	590+
Timeout	0	0
Average solve time	42 ms	40 ms

Metric	Value
Minimum $\mathcal{E}_{\text{rev}}$ found	60
Required for success	0
Gap	≥ 60

The consistent gap of ≥ 60 energy units across all 18 trials (using different near-miss L_3^* seeds) strongly indicates that the near-miss L_3 matrices are not structurally compatible with any MOLS-10 pair.

4.7 Local Minimum Analysis

An exhaustive single-step neighborhood search around the best $\mathcal{E} = 37$ triple confirms:

Move type	Neighborhood	Fraction improving	Min $\Delta\mathcal{E}$
Row/col/relabel	$3 \times 45 = 135$	0%	+2
Intercalate	$\approx 7,500$	0%	+2
kscramble ($k = 3$)	≈ 720	0%	+2
bigscramble ($k = 6$)	$\approx 5,040$	0%	+4

$\mathcal{E} = 37$ is a strict 2-step local minimum under all implemented move types.

5 Near-Miss Triple Analysis

5.1 Pool State

Table 7 shows the current near-miss pool (top 8 triples by energy).

Table 7: Near-miss pool state (Session 7).

Index	$\mathcal{E}$	cl_{12}	cl_{13}	cl_{23}
0	37	0	22	15
1	37	0	22	15
2	37	0	22	15
3	37	0	22	15
4	37	0	22	15
5	43	0	22	21
6	43	0	21	22
7	43	0	22	21

All five $\mathcal{E} = 37$ entries have the same clash breakdown $(\text{cl}_{13}, \text{cl}_{23}) = (22, 15)$, suggesting a structural constraint on the minimum achievable clashes for a pair with CT = 2.

5.2 L3 Diversity

The five L_3 matrices in the $\mathcal{E} = 37$ pool are distinct, with pairwise Hamming distances:

	$L_3^{(0)}$	$L_3^{(1)}$	$L_3^{(2)}$	$L_3^{(3)}$	$L_3^{(4)}$
$L_3^{(0)}$	0	60	80	80	70
$L_3^{(1)}$	60	0	20	30	80
$L_3^{(2)}$	80	20	0	40	90
$L_3^{(3)}$	80	30	40	0	80
$L_3^{(4)}$	70	80	90	80	0

(Values out of 100 cells.) The maximum distance of 90/100 (between $L_3^{(2)}$ and $L_3^{(4)}$) confirms the search reaches genuinely different regions of the L_3 space, all achieving the same minimum energy.

5.3 Statistical Interpretation of E=37

Proposition 5.1. *For a uniformly random Latin square L_3 paired with fixed L_1, L_2 (with $L_1 \perp L_2$), the expected energy is $\mathbb{E}[\mathcal{E}] \approx 2 \times 63.2 = 126.4$ (using Lemma 2.5).*

Our SA descends from this random baseline ($\mathcal{E} \approx 126$) to $\mathcal{E} = 37$ — a reduction of $\approx 70\%$ — representing substantial combinatorial optimization. Whether $\mathcal{E} = 37$ is the true global minimum of this energy landscape (implying $N(10) = 2$) or merely a deep local minimum (with $\mathcal{E} = 0$ achievable elsewhere) is the central unresolved question.

6 The CT Barrier

6.1 Empirical Finding

Key empirical finding: Every MOLS-10 pair (L_1, L_2) generated by this project satisfies $\text{CT}(L_1, L_2) \leq 2$.

This holds across:

- 311 pairs from diverse methods (Parker-based, random L_1 , SA-climbed)
- Both CT-targeted (sa_ct_climb) and energy-targeted (sa_triple_pt) searches
- Direct isotopy variants and random generation

Since $\text{CT} < 10 = n$ for all pairs, Corollary 2.3 applies: **no third orthogonal square exists for any tested pair.**

6.2 Parker TV-21 Analysis

The canonical Parker (1960) pair has $\text{CT} = 2$, verified by exhaustive enumeration. We additionally tested 12 distinct orthogonal mates of Parker’s L_1 (generated by multi-decomposition), all yielding $\text{CT} = 2$.

6.3 Algebraic Interpretation

For Galois field order q , MOLS pairs from the standard construction have $\text{CT} = q$. The absence of $GF(10)$ means this algebraic route is unavailable. The available algebraic groups of order 10 ($\mathbb{Z}_{10}$ and D_5) produce Latin squares (via Cayley tables) that we have tested with full isotopy; none achieve $\text{CT} > 2$.

6.4 Conjectures

Conjecture 6.1 (CT Universality). For all MOLS-10 pairs (L_1, L_2) , $\text{CT}(L_1, L_2) \leq 2$.

Conjecture 6.1, if true, would immediately imply $N(10) = 2$ via Corollary 2.3. Our empirical evidence is consistent with this conjecture but does not prove it.

Conjecture 6.2 (Energy Floor). For any Latin squares L_1, L_2, L_3 of order 10 with $L_1 \perp L_2$: $\mathcal{E}(L_1, L_2, L_3) \geq 37$.

7 $N(10) \geq 3$: Empirical Evidence Summary

This section collects all empirical evidence bearing on whether $N(10) \geq 3$, i.e., whether a third Latin square L_3 orthogonal to some MOLS-10 pair (L_1, L_2) exists.

7.1 Evidence Against $N(10) \geq 3$

1. CT barrier (strongest evidence). By the CT Necessity Theorem (Theorem 2.2), any $L_3 \perp L_1, L_3 \perp L_2$ requires $\text{CT}(L_1, L_2) \geq 10$. Exhaustive backtracking over all 311 tested pairs confirms:

$$\max_{(L_1, L_2) \in \text{catalog}} \text{CT}(L_1, L_2) = 2 \ll 10.$$

No tested pair can support an orthogonal mate L_3 .

2. SAT certificates (definitive per-pair proofs). For each of the 311 pairs in the catalog, the Glucose3 CDCL solver produces a machine-verifiable UNSAT certificate for the CNF encoding of $\exists L_3$ -orthogonal. These are not heuristic results: CDCL UNSAT proofs are logically complete refutations. Additionally, ≈ 600 independently generated pairs have been SAT-certified UNSAT. No SAT instance has ever returned satisfiable (L_3 found).

3. Energy barrier $E = 37$. Five sessions of algorithmic improvements have not breached $\mathcal{E} = 37$ despite: 7-replica PT-SA, pool-jump ILS, 11 initialization strategies, and $> 10^9$ total SA steps. The barrier is a strict 2-step local minimum under all implemented move types.

4. Reverse search failure. Fixing the best near-miss L_3 and running 18 independent reverse-search trials (7-replica PT over (L_1, L_2)) consistently yields $\mathcal{E}_{\text{rev}} \geq 60$. The gap of ≥ 60 energy units indicates the near-miss L_3 matrices are structurally incompatible with any MOLS-10 pair.

7.2 Evidence For $N(10) \geq 3$ (Currently None)

No positive evidence for $N(10) \geq 3$ has been found:

- No pair with $\text{CT} \geq 3$ has been found.
- No SAT instance has returned a satisfying assignment.
- No energy below 37 has been achieved for any triple with $\text{cl}_{12} = 0$.
- Reverse search has never found a triple with $\mathcal{E}_{\text{rev}} < 60$.

7.3 Interpretation

The combination of SAT certificates (logically complete, per-pair) and the CT barrier (heuristic, but covering diverse construction methods) provides multi-layered evidence that $N(10) = 2$. The CT barrier, if it holds universally (Conjecture 6.1), would constitute a proof; the SAT certificates prove it for each tested pair individually.

The $\mathcal{E} = 37$ near-miss triples are the best known approximations to a 3-MOLS triple and may be useful for future theoretical or computational attacks, but currently provide no path toward $N(10) \geq 3$.

8 Conclusions and Future Work

8.1 Main Conclusion

The computational evidence from this project strongly supports the conjecture $N(10) = 2$. We have found no three mutually orthogonal Latin squares of order 10.

Supporting evidence:

- **CT barrier:** $\text{CT} \leq 2$ for all 311 tested MOLS-10 pairs, proven by exhaustive enumeration.
- **SAT certificates:** Machine-verifiable UNSAT proofs for all 311 pairs (42 ms average) and 600+ additional pairs; zero SAT results.
- **$\mathcal{E}=37$ barrier:** Three consecutive sessions of improvements have not breached $\mathcal{E} = 37$; this is a strict 2-step local minimum under 6 move types.
- **Reverse search:** Near-miss L_3 matrices are incompatible with any MOLS structure ($\mathcal{E}_{\text{rev}} \geq 60$ across 18 trials).

8.2 Open Questions

1. Is Conjecture 6.1 provable? A proof would resolve $N(10)$ definitively.
2. What is the true minimum of $\mathcal{E}$ over all triples with $\text{cl}_{12} = 0$? Our empirical minimum is 37; can $\mathcal{E} < 37$ be achieved?
3. Are the four $\mathcal{E} = 37$ near-miss triples in distinct isotopy classes?
4. Does the CT distribution (always ≤ 2) follow from a group-theoretic argument related to the factorization $10 = 2 \times 5$?

8.3 Future Directions

1. **Wider CT search** with >10,000 pairs to test Conjecture 6.1.
2. **MAX-SAT minimization** to find the true minimum energy for specific pairs.
3. **Isotopy classification** of the near-miss L_3 matrices.
4. **Larger compute scale** (50+ workers on HPC cluster) for broader isotopy class coverage.
5. **Intercalate theory analysis** relating CT count to intercalate structure of MOLS-10 pairs (following Wanless & Webb [12]).

A Representative MOLS-10 Pairs with Verification

This appendix gives full 10×10 matrices for three representative pairs from the catalog. For each pair, orthogonality is confirmed by verifying $\text{cl}(L_1, L_2) = 0$. Complete data for all 311 pairs is in `promising_pairs.json`.

A.1 Parker TV-21 (1960) — CT = 2

The canonical pair introduced by Parker [10]. This is the reference pair from which all **tv** and **ic*** variants derive. Verification: both squares are valid Latin squares; all 100 ordered symbol pairs appear exactly once in the superposition; $\text{cl}(L_1, L_2) = 0$; $\text{CT}(L_1, L_2) = 2$ (verified by exhaustive backtracking).

$$L_1^{\text{TV21}} = \begin{pmatrix} 0 & 9 & 4 & 7 & 6 & 5 & 3 & 8 & 2 & 1 \\ 9 & 5 & 0 & 2 & 3 & 8 & 6 & 4 & 1 & 7 \\ 2 & 6 & 9 & 0 & 4 & 3 & 1 & 7 & 5 & 8 \\ 4 & 1 & 6 & 9 & 8 & 2 & 7 & 0 & 3 & 5 \\ 7 & 8 & 5 & 1 & 2 & 0 & 4 & 9 & 6 & 3 \\ 1 & 7 & 3 & 4 & 5 & 6 & 8 & 2 & 0 & 9 \\ 8 & 3 & 7 & 5 & 1 & 9 & 2 & 6 & 4 & 0 \\ 6 & 0 & 2 & 3 & 9 & 7 & 5 & 1 & 8 & 4 \\ 5 & 4 & 8 & 6 & 7 & 1 & 0 & 3 & 9 & 2 \\ 3 & 2 & 1 & 8 & 0 & 4 & 9 & 5 & 7 & 6 \end{pmatrix} \quad L_2^{\text{TV21}} = \begin{pmatrix} 7 & 4 & 8 & 6 & 2 & 3 & 9 & 5 & 0 & 1 \\ 2 & 7 & 3 & 9 & 1 & 0 & 5 & 4 & 6 & 8 \\ 3 & 0 & 7 & 5 & 9 & 8 & 4 & 2 & 1 & 6 \\ 0 & 3 & 9 & 8 & 4 & 1 & 7 & 6 & 5 & 2 \\ 1 & 9 & 6 & 0 & 5 & 4 & 2 & 3 & 8 & 7 \\ 9 & 5 & 4 & 1 & 8 & 6 & 3 & 7 & 2 & 0 \\ 8 & 2 & 0 & 4 & 7 & 5 & 6 & 1 & 3 & 9 \\ 4 & 1 & 2 & 3 & 6 & 9 & 0 & 8 & 7 & 5 \\ 5 & 6 & 1 & 7 & 3 & 2 & 8 & 0 & 9 & 4 \\ 6 & 8 & 5 & 2 & 0 & 7 & 1 & 9 & 4 & 3 \end{pmatrix}$$

Orthogonality check: Superimposing L_1 and L_2 yields 100 cells; each of the 100 ordered pairs $(a, b) \in \{0, \dots, 9\}^2$ appears exactly once. $\text{cl}(L_1^{\text{TV21}}, L_2^{\text{TV21}}) = 0 \checkmark$.

A.2 TV-10 Variant — CT = 1 (new finding)

A transversal variant of the Parker L_1 , with a different orthogonal mate constructed by the **tv** method. This pair achieves $\text{CT} = 1$: there exists exactly one common transversal, which cannot form a partition of all 100 cells (10 disjoint CTs required for any valid L_3). By Corollary 2.3, no L_3 orthogonal to both squares exists. Verification: $\text{cl}(L_1, L_2) = 0$; $\text{CT} = 1$ (exhaustive enumeration).

$$L_1^{\text{TV10}} = \begin{pmatrix} 4 & 3 & 1 & 7 & 5 & 6 & 9 & 8 & 2 & 0 \\ 7 & 8 & 0 & 6 & 1 & 2 & 4 & 9 & 3 & 5 \\ 8 & 6 & 4 & 1 & 9 & 5 & 3 & 7 & 0 & 2 \\ 5 & 9 & 6 & 3 & 4 & 0 & 1 & 2 & 7 & 8 \\ 2 & 0 & 7 & 5 & 6 & 4 & 8 & 1 & 9 & 3 \\ 1 & 7 & 9 & 0 & 8 & 3 & 2 & 5 & 6 & 4 \\ 9 & 5 & 8 & 4 & 2 & 7 & 0 & 3 & 1 & 6 \\ 6 & 2 & 3 & 9 & 0 & 1 & 5 & 4 & 8 & 7 \\ 3 & 4 & 2 & 8 & 7 & 9 & 6 & 0 & 5 & 1 \\ 0 & 1 & 5 & 2 & 3 & 8 & 7 & 6 & 4 & 9 \end{pmatrix} \quad L_2^{\text{TV10}} = \begin{pmatrix} 3 & 8 & 0 & 5 & 1 & 2 & 4 & 7 & 9 & 6 \\ 9 & 4 & 5 & 1 & 8 & 6 & 7 & 2 & 3 & 0 \\ 2 & 0 & 4 & 7 & 3 & 9 & 5 & 6 & 1 & 8 \\ 4 & 5 & 6 & 2 & 0 & 7 & 9 & 1 & 8 & 3 \\ 0 & 9 & 2 & 3 & 4 & 8 & 1 & 5 & 6 & 7 \\ 1 & 3 & 9 & 0 & 6 & 4 & 2 & 8 & 7 & 5 \\ 7 & 2 & 8 & 6 & 5 & 1 & 3 & 0 & 4 & 9 \\ 5 & 7 & 1 & 8 & 2 & 3 & 6 & 9 & 0 & 4 \\ 6 & 1 & 3 & 9 & 7 & 0 & 8 & 4 & 5 & 2 \\ 8 & 6 & 7 & 4 & 9 & 5 & 0 & 3 & 2 & 1 \end{pmatrix}$$

Note: $\text{CT} = 1$ is the minimum possible for a MOLS pair. The existence of such pairs with $\text{CT} = 1$ is itself a new finding; prior to this work no systematic CT enumeration of MOLS-10 pairs had been published.

A.3 Transversal Decomposition Pair mdecomp_48 — CT = 2 (new construction)

Generated by the **mdecomp** method: a random Latin square L_1 is constructed (not isotopic to Parker's), then a transversal decomposition finds an orthogonal mate L_2 . This pair and all 123 other **mdecomp** pairs have L_1 matrices outside the Parker isotopy class, making them genuinely independent new MOLS-10 pairs. Verification: $\text{cl}(L_1, L_2) = 0$; $\text{CT} = 2$.

$$L_1^{\text{md48}} = \begin{pmatrix} 8 & 1 & 4 & 2 & 6 & 3 & 0 & 5 & 9 & 7 \\ 4 & 6 & 9 & 0 & 8 & 5 & 2 & 7 & 1 & 3 \\ 3 & 7 & 5 & 6 & 2 & 4 & 8 & 9 & 0 & 1 \\ 5 & 9 & 1 & 3 & 7 & 8 & 6 & 0 & 2 & 4 \\ 7 & 8 & 0 & 4 & 1 & 2 & 9 & 3 & 5 & 6 \\ 9 & 5 & 2 & 8 & 3 & 1 & 4 & 6 & 7 & 0 \\ 6 & 4 & 3 & 1 & 9 & 0 & 7 & 2 & 8 & 5 \\ 2 & 3 & 7 & 5 & 0 & 6 & 1 & 8 & 4 & 9 \\ 0 & 2 & 6 & 7 & 4 & 9 & 5 & 1 & 3 & 8 \\ 1 & 0 & 8 & 9 & 5 & 7 & 3 & 4 & 6 & 2 \end{pmatrix} \quad L_2^{\text{md48}} = \begin{pmatrix} 8 & 9 & 5 & 2 & 0 & 4 & 1 & 6 & 3 & 7 \\ 9 & 2 & 8 & 7 & 3 & 1 & 5 & 0 & 4 & 6 \\ 2 & 8 & 7 & 3 & 6 & 0 & 4 & 5 & 9 & 1 \\ 3 & 6 & 0 & 8 & 1 & 5 & 9 & 2 & 7 & 4 \\ 4 & 7 & 6 & 1 & 8 & 3 & 0 & 9 & 2 & 5 \\ 1 & 4 & 9 & 0 & 7 & 2 & 6 & 8 & 5 & 3 \\ 7 & 3 & 1 & 5 & 9 & 8 & 2 & 4 & 6 & 0 \\ 0 & 5 & 3 & 9 & 4 & 6 & 7 & 1 & 8 & 2 \\ 5 & 1 & 4 & 6 & 2 & 7 & 8 & 3 & 0 & 9 \\ 6 & 0 & 2 & 4 & 5 & 9 & 3 & 7 & 1 & 8 \end{pmatrix}$$

Orthogonality check: $\text{cl}(L_1^{\text{md48}}, L_2^{\text{md48}}) = 0 \checkmark$. $\text{CT} = 2$. This pair's L_1 has first row $[8, 1, 4, 2, 6, 3, 0, 5, 9, 7]$, confirming it is not isotopic to TV-21's L_1 (which has first row $[0, 9, 4, 7, 6, 5, 3, 8, 2, 1]$ in reduced form).

A.4 Complete 311-Pair Summary Table

Construction	Total	CT=1	CT=2
tv (Parker transversal variants)	56	46	10
ic2 (intercalate chain, depth 2)	3	1	2
ic3 (intercalate chain, depth 3)	3	1	2
ic5 (intercalate chain, depth 5)	5	2	3
ic8 (intercalate chain, depth 8)	4	1	3
ic12 (intercalate chain, depth 12)	3	0	3
ic1.3 (L1 intercalate, depth 3)	2	1	1
ic1.5 (L1 intercalate, depth 5)	2	1	1
iso (isotopy transforms)	6	3	3
mdecomp (transversal decomposition)	124	2	122
sa_ct (SA CT-climbing)	103	0	103
Total	311	58	253

All 311 pairs are verified MOLS ($\text{cl} = 0$) with CT computed by exhaustive backtracking. **No pair has** $\text{CT} > 2$ — every pair is SAT-certified to have no orthogonal mate L_3 (Glucose3 CDCL, average 42 ms/pair, 0 timeouts).

B Data Tables

Sample E=37 Near-Miss Triple

One of the five $\mathcal{E} = 37$ triples (pool entry 0, found 2026-06-07 18:41). All matrices have symbols $\{0, \dots, 9\}$.

The full triple $(L_1^{(0)}, L_2^{(0)}, L_3^{(0)})$ from pool entry 0. $L_1^{(0)} \perp L_2^{(0)}$ is verified ($\text{cl}_{12} = 0$); $L_3^{(0)}$ is a valid Latin square with $\text{cl}_{13} = 22$, $\text{cl}_{23} = 15$.

$$L_1^{(0)} = \begin{pmatrix} 4 & 7 & 3 & 2 & 8 & 9 & 5 & 1 & 0 & 6 \\ 8 & 2 & 5 & 1 & 0 & 4 & 7 & 6 & 3 & 9 \\ 1 & 0 & 9 & 3 & 4 & 8 & 6 & 2 & 7 & 5 \\ 7 & 3 & 6 & 4 & 5 & 0 & 8 & 9 & 2 & 1 \\ 9 & 8 & 4 & 7 & 6 & 1 & 2 & 0 & 5 & 3 \\ 3 & 1 & 0 & 5 & 7 & 6 & 9 & 8 & 4 & 2 \\ 0 & 4 & 1 & 6 & 2 & 5 & 3 & 7 & 9 & 8 \\ 6 & 5 & 2 & 8 & 9 & 7 & 0 & 3 & 1 & 4 \\ 5 & 6 & 7 & 9 & 3 & 2 & 1 & 4 & 8 & 0 \\ 2 & 9 & 8 & 0 & 1 & 3 & 4 & 5 & 6 & 7 \end{pmatrix} \quad L_2^{(0)} = \begin{pmatrix} 6 & 0 & 2 & 3 & 5 & 9 & 7 & 1 & 4 & 8 \\ 3 & 4 & 6 & 7 & 8 & 5 & 1 & 0 & 9 & 2 \\ 2 & 3 & 5 & 1 & 0 & 6 & 9 & 8 & 7 & 4 \\ 9 & 8 & 7 & 4 & 2 & 1 & 0 & 3 & 5 & 6 \\ 8 & 7 & 9 & 2 & 1 & 4 & 6 & 5 & 3 & 0 \\ 5 & 9 & 0 & 8 & 6 & 3 & 4 & 2 & 1 & 7 \\ 7 & 2 & 8 & 5 & 9 & 0 & 3 & 4 & 6 & 1 \\ 4 & 5 & 1 & 9 & 7 & 8 & 2 & 6 & 0 & 3 \\ 1 & 6 & 3 & 0 & 4 & 2 & 5 & 7 & 8 & 9 \\ 0 & 1 & 4 & 6 & 3 & 7 & 8 & 9 & 2 & 5 \end{pmatrix}$$

$$L_3^{(0)} = \begin{pmatrix} 6 & 1 & 2 & 0 & 5 & 7 & 3 & 9 & 4 & 8 \\ 8 & 9 & 0 & 1 & 6 & 3 & 2 & 7 & 5 & 4 \\ 4 & 2 & 3 & 7 & 8 & 1 & 0 & 5 & 6 & 9 \\ 9 & 3 & 7 & 5 & 1 & 6 & 4 & 8 & 2 & 0 \\ 2 & 0 & 5 & 8 & 3 & 4 & 7 & 1 & 9 & 6 \\ 3 & 8 & 9 & 4 & 7 & 5 & 1 & 6 & 0 & 2 \\ 0 & 7 & 1 & 9 & 4 & 2 & 6 & 3 & 8 & 5 \\ 7 & 6 & 8 & 2 & 9 & 0 & 5 & 4 & 3 & 1 \\ 1 & 5 & 4 & 6 & 0 & 9 & 8 & 2 & 7 & 3 \\ 5 & 4 & 6 & 3 & 2 & 8 & 9 & 0 & 1 & 7 \end{pmatrix}$$

Verified: $L_1^{(0)} \perp L_2^{(0)}$ ($\text{cl}_{12} = 0$); $L_3^{(0)}$ is a valid Latin square; $\text{cl}_{13} = 22$, $\text{cl}_{23} = 15$, $\mathcal{E} = 37$. Full data for all five $\mathcal{E} = 37$ triples is in `mols10/results/near_miss_triple.json`.

Notation Summary

Symbol	Meaning
$N(n)$	Max number of MOLS of order n
L_i	Latin square i (10×10 , symbols 0–9)
$L_i \perp L_j$	L_i orthogonal to L_j
$\text{cl}(A, B)$	Clash count: symbol pairs missing from superposition
$\mathcal{E}$	Total energy = $\text{cl}_{12} + \text{cl}_{13} + \text{cl}_{23}$
$\text{CT}(L_1, L_2)$	Common transversal count
SA	Simulated annealing
PT	Parallel tempering
ILS	Iterated local search
CDCL	Conflict-driven clause learning

Data availability. All near-miss triples, MOLS pairs, worker logs, and search code are available at <https://github.com/ajzhanghk/autoresearch-glm>, branch `claude/mols-order-10-search-yfQXK`.

References

- [1] Audemard, G., & Simon, L. (2009). Predicting learnt clauses quality in modern SAT solvers. *Proceedings of IJCAI*, 399–404.
- [2] Clements, G. F., & Lindström, B. (1965). A generalization of a combinatorial theorem of Macaulay. *Journal of Combinatorial Theory*, 7(3), 230–238.
- [3] Colbourn, C. J., & Dinitz, J. H. (Eds.) (2006). *Handbook of Combinatorial Designs* (2nd ed.). Chapman & Hall/CRC.
- [4] Dénes, J., & Keedwell, A. D. (1991). *Latin Squares: New Developments in the Theory and Applications*. North-Holland.
- [5] Euler, L. (1782). Recherches sur une nouvelle espèce de quarrés magiques. *Verhandelingen uitgegeven door het zeeuwsch genootschap der wetenschappen te Vlissingen*, 9, 85–239.
- [6] Geyer, C. J. (1991). Markov chain Monte Carlo maximum likelihood. *Proceedings of the 23rd Symposium on the Interface*, 156–163.
- [7] Knuth, D. E. (2000). Dancing links. *Millennial Perspectives in Computer Science*, 187–214.
- [8] Lam, C. W. H., Thiel, L., & Swiercz, S. (1989). The nonexistence of finite projective planes of order 10. *Canadian Journal of Mathematics*, 41(6), 1117–1123.

- [9] McKay, B. D., Meynert, A., & Myrvold, W. (2007). Small Latin squares, quasigroups, and loops. *Journal of Combinatorial Designs*, 15(2), 98–119.
- [10] Parker, E. T. (1960). Orthogonal Latin squares. *Proceedings of the National Academy of Sciences*, 47(6), 859–862.
- [11] Tarry, G. (1901). Le problème des 36 officiers. *Compte Rendu de l'Association Française pour l'Avancement des Sciences Naturelles*, 2, 170–203.
- [12] Wanless, I. M., & Webb, B. S. (2011). The existence of Latin squares without orthogonal mates. *Designs, Codes and Cryptography*, 60(2), 143–151.